Will Vance

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Summary

Computer Science undergraduate at CU Boulder building real-time software across computer vision, audio plugins, and web development in Python, C++, and JavaScript. Seeking a Summer 2027 software engineering internship.

Skills

Programming Languages: C, C++, Python, JavaScript, SQL, HTML/CSS, Bash
Frameworks/Libraries: React, Vite, OpenCV, MediaPipe, JUCE, Blender Python API (bpy)
Tools: Git, Docker, PostgreSQL, Linux, Cloudflare Pages, VSCode, Visual Studio, WSL2, Arduino
AI Tools: Agentic coding workflows with Claude Code

Projects

Body Tracker – Motion Capture (Python, OpenCV, MediaPipe, Blender API) *Fall 2026 – Present*

- Built a real-time motion-capture pipeline in Python that tracks up to 46 hand and body landmarks from a webcam with MediaPipe and streams them to a Blender add-on that drives a 47-bone rig live
- Implemented a One Euro filter to smooth noisy landmark data, cutting jitter ∼35% versus a moving average while reducing lag from 163 ms to 31 ms
- Designed a versioned JSON-over-UDP protocol with sequence numbers and timestamps (≤2.6 KB/frame) plus a listener that detects lost or out-of-order packets, sustaining ∼28 packets/s with zero loss locally
- Developed a Blender extension that polls a non-blocking socket from Blender's main thread (bpy is not thread-safe) and generates an armature with bone constraints programmatically

Personal Website – wjvance.com (React, Vite, JavaScript, CSS, Cloudflare) *Fall 2026*

- Built a responsive single-page portfolio in React 19 and Vite, rebuilt from an earlier plain-HTML class site, with continuous deployment through Cloudflare on every push to main
- Wrote a reusable scroll-reveal component using the IntersectionObserver API with staggered CSS transitions and reduced-motion support
- Kept all site content in a single data module and the theme in CSS custom-property design tokens, so copy, links, and colors update without touching component code
- Kept the bundle to ∼65 KB gzipped JS by using dependency-free inline SVG icons and replacing a demo GIF with a ∼425 KB looping MP4

SquillysDelay – Audio Plugin (C++, JUCE) *Fall 2026 – Present*

- Building a tape-echo style delay effect as a VST3 plugin in C++ with the JUCE framework, targeting FL Studio
- Designing a real-time DSP signal chain (delay line, feedback, wet/dry mix) under low-latency audio constraints

Education

University of Colorado Boulder — GPA: 3.19 *Expected Graduation: Spring 2028*
Bachelor of Science in Computer Science
Relevant Coursework: Data Structures, Computer Systems, Software Development, Discrete Mathematics, Data Science, Algorithms

Work Experience

Rocky Mountain Chocolate Factory *2022–2024*
Sales Associate

- Handled POS transactions, cash, and inventory in a fast-paced retail and food-service environment

Action Behavior Centers *June 2024–Oct 2024*
Registered Behavioral Therapist

- Delivered structured behavioral interventions for children with Autism Spectrum Disorder in a clinical setting
- Documented session data in HIPAA-compliant software and coordinated care plans with board-certified behavior analysts